

BEYOND BASIC RAG

The 4-step architecture blueprint for audit-ready, zero-trust enterprise AI systems.

The Production Trust Gap

Building a RAG demo is simple. Moving it to production for financial compliance or auditing exposes major flaws:

1

PII Ingestion Risks

Feeding raw, unredacted corporate documents directly to public LLMs violates data privacy laws (GDPR, HIPAA).

2

Context Fragmentation

Basic token-based chunking slices sentences in half, causing the model to lose context on complex tables or legal definitions.

3

Missing Citations

Models generate answers but cannot trace their statements back to the exact page, clause, or paragraph of the source document.

1. Zero-Trust Ingestion

Never ingest raw data. We sanitize files at the entry point using structural parsing and named entity recognition:

A

Structure Extraction

Use tools like Azure Document Intelligence to parse headers, tables, and sections while keeping layouts intact.

B

PII Redaction Pipeline

Run raw text through a high-performance NER pipeline. Scrub names, tax identifiers, and emails before saving them to the vector store.



Raw Docs



Azure DI



NER Scrub



Vector DB

2. Precision Citations

Avoid generic matches. Implement a metadata-rich parent-child document chunking strategy that forces LLMs to reference source locations:

1

Parent-Child Chunking

Index small sub-chunks (100 tokens) for precise search matches, but retrieve their larger parent contexts (500-1000 tokens) to supply rich context to the LLM.

2

High-Fidelity Metadata Mapping

Inject metadata tags directly into each chunk. Store document keys, precise page coordinates, clause IDs, and exact verbatim snippets.

3

Enforced Citation Anchors

Format prompts to require the LLM to output footnotes. If the final output states a fact, it must link back to an index key.

3. Agentic Verification

Linear pipelines are brittle. Instead, build stateful agent loops using LangGraph to plan, search, check, and optimize replies:

1

The Planner Agent

Deconstructs complex user prompts into multi-step execution tasks (e.g., executing separate vector queries, SQL views, and comparison steps).

2

The Retrieval Validator

Executes hybrid search (Keyword BM25 + Vector) and confirms retrieved documents contain the exact metadata required for verification.

3

The Brand & Safety Guard

A separate LLM agent acts as a gating mechanism. It reviews the final output against hallucinations and strict brand-safety parameters.

The Enterprise Tech Stack

Putting the architectural pieces together with industry-leading frameworks:

LG

LangGraph

Stateful Agent Loops

PC

Pinecone / Azure

Hybrid Vector Search

AZ

Azure OpenAI

GPT-4o & Embeddings

MP

Model Context

Conversational BI / SQL

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